

Instituto Tecnológico de Costa Rica

Operations Research - Semester II

Simplex

Members:

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Non Feasible

Maximize

$$Z = 300.00x_1 + 200.00x_2$$

Subject to

$$0.03x_1 + 0.02x_2 \leq 1.00$$

$$0.02x_1 + 0.02x_2 \leq 1.00$$

$$1.00x_1 + 0.00x_2 \geq 30.00$$

$$0.00x_1 + 1.00x_2 \geq 20.00$$

Simplex Table

	Z	x_1	x_2	s_1	s_2	e_1	e_2	a_1	a_2	b
	1	-300	-200	0	0	0	0	M	M	0
	0	0.025	0.017	1	0	0	0	0	0	1
	0	0.02	0.02	0	1	0	0	0	0	1
	0	1	0	0	0	-1	0	1	0	30
	0	0	1	0	0	0	-1	0	1	20

Artificial Variables "Canonization"

Variable a_1

$$R_1 \leftarrow R_1 + -30MR_4$$

	Z	x_1	x_2	s_1	s_2	e_1	e_2	a_1	a_2	b
	1	-1M	-200	0	0	M	0	0	M	-30M
	0	0.025	0.017	1	0	0	0	0	0	1
	0	0.02	0.02	0	1	0	0	0	0	1
	0	1	0	0	0	-1	0	1	0	30
	0	0	1	0	0	0	-1	0	1	20

Variable a_2

$$R_1 \leftarrow R_1 + -20MR_5$$

	Z	x_1	x_2	s_1	s_2	e_1	e_2	a_1	a_2	b
	1	-1M	-1M	0	0	M	M	0	0	-50M
	0	0.025	0.017	1	0	0	0	0	0	1
	0	0.02	0.02	0	1	0	0	0	0	1
	0	1	0	0	0	-1	0	1	0	30
	0	0	1	0	0	0	-1	0	1	20

Intermediate Tables

Pivoting 1

Most Negative

Column 2 (-1M)

	Z	x_1	x_2	s_1	s_2	e_1	e_2	a_1	a_2	b
	1	-1M	-1M	0	0	M	M	0	0	-50M
	0	0.025	0.017	1	0	0	0	0	0	1
	0	0.02	0.02	0	1	0	0	0	0	1
	0	1	0	0	0	-1	0	1	0	30
	0	0	1	0	0	0	-1	0	1	20

Fractions

	Z	x_1	x_2	s_1	s_2	e_1	e_2	a_1	a_2	b	Frac
	1	-1M	-1M	0	0	M	M	0	0	-50M	
	0	0.025	0.017	1	0	0	0	0	0	1	40.00
	0	0.02	0.02	0	1	0	0	0	0	1	50.00
	0	1	0	0	0	-1	0	1	0	30	30.00
	0	0	1	0	0	0	-1	0	1	20	inf

$$1.00/0.03 = 40.00$$

$$1.00/0.02 = 50.00$$

$$30.00/1.00 = 30.00$$

$$20.00/0.00 = \text{inf}$$

Smallest fraction: 30.00 $\rightarrow$ Pivot: row 4

Pivot

	Z	x_1	x_2	s_1	s_2	e_1	e_2	a_1	a_2	b
	1	-1M	-1M	0	0	M	M	0	0	-50M
	0	0.025	0.017	1	0	0	0	0	0	1
	0	0.02	0.02	0	1	0	0	0	0	1
	0	1	0	0	0	-1	0	1	0	30
	0	0	1	0	0	0	-1	0	1	20

Canonization

$$R_4 \leftarrow R_4/1.00$$

	Z	x_1	x_2	s_1	s_2	e_1	e_2	a_1	a_2	b
	1	-1M	-1M	0	0	M	M	0	0	-50M
	0	0.025	0.017	1	0	0	0	0	0	1
	0	0.02	0.02	0	1	0	0	0	0	1
	0	1	0	0	0	-1	0	1	0	30
	0	0	1	0	0	0	-1	0	1	20

$$R_1 \leftarrow R_1 + 1000300.00R_4$$

	Z	x_1	x_2	s_1	s_2	e_1	e_2	a_1	a_2	b
	1	0	-1M	0	0	-300	M	1M	0	-20M
	0	0.025	0.017	1	0	0	0	0	0	1
	0	0.02	0.02	0	1	0	0	0	0	1
	0	1	0	0	0	-1	0	1	0	30
	0	0	1	0	0	0	-1	0	1	20

$$R_2 \leftarrow R_2 + -0.03R_4$$

	Z	x_1	x_2	s_1	s_2	e_1	e_2	a_1	a_2	b
	1	0	-1M	0	0	-300	M	1M	0	-20M
	0	0	0.017	1	0	0.025	0	-0.025	0	0.25
	0	0.02	0.02	0	1	0	0	0	0	1
	0	1	0	0	0	-1	0	1	0	30
	0	0	1	0	0	0	-1	0	1	20

$$R_3 \leftarrow R_3 + -0.02R_4$$

	Z	x_1	x_2	s_1	s_2	e_1	e_2	a_1	a_2	b
	1	0	-1M	0	0	-300	M	1M	0	-20M
	0	0	0.017	1	0	0.025	0	-0.025	0	0.25
	0	0	0.02	0	1	0.02	0	-0.02	0	0.4
	0	1	0	0	0	-1	0	1	0	30
	0	0	1	0	0	0	-1	0	1	20

$$R_5 \leftarrow R_5 + -0.00R_4$$

	Z	x_1	x_2	s_1	s_2	e_1	e_2	a_1	a_2	b
	1	0	-1M	0	0	-300	M	1M	0	-20M
	0	0	0.017	1	0	0.025	0	-0.025	0	0.25
	0	0	0.02	0	1	0.02	0	-0.02	0	0.4
	0	1	0	0	0	-1	0	1	0	30
	0	0	1	0	0	0	-1	0	1	20

Pivot Result

	Z	x_1	x_2	s_1	s_2	e_1	e_2	a_1	a_2	b
	1	0	-1M	0	0	-300	M	1M	0	-20M
	0	0	0.017	1	0	0.025	0	-0.025	0	0.25
	0	0	0.02	0	1	0.02	0	-0.02	0	0.4
	0	1	0	0	0	-1	0	1	0	30
	0	0	1	0	0	0	-1	0	1	20

Pivoting 2

Most Negative

Column 3 (-1M)

	Z	x_1	x_2	s_1	s_2	e_1	e_2	a_1	a_2	b
	1	0	-1M	0	0	-300	M	1M	0	-20M
	0	0	0.017	1	0	0.025	0	-0.025	0	0.25
	0	0	0.02	0	1	0.02	0	-0.02	0	0.4
	0	1	0	0	0	-1	0	1	0	30
	0	0	1	0	0	0	-1	0	1	20

Fractions

	Z	x_1	x_2	s_1	s_2	e_1	e_2	a_1	a_2	b	Frac
	1	0	-1M	0	0	-300	M	1M	0	-20M	
	0	0	0.017	1	0	0.025	0	-0.025	0	0.25	14.71
	0	0	0.02	0	1	0.02	0	-0.02	0	0.4	20.00
	0	1	0	0	0	-1	0	1	0	30	inf
	0	0	1	0	0	0	-1	0	1	20	20.00

$$0.25/0.02 = 14.71$$

$$0.40/0.02 = 20.00$$

$$30.00/0.00 = \text{inf}$$

$$20.00/1.00 = 20.00$$

Smallest fraction: 14.71 $\rightarrow$ Pivot: row 2

Pivot

	Z	x_1	x_2	s_1	s_2	e_1	e_2	a_1	a_2	b
	1	0	-1M	0	0	-300	M	1M	0	-20M
	0	0	0.017	1	0	0.025	0	-0.025	0	0.25
	0	0	0.02	0	1	0.02	0	-0.02	0	0.4
	0	1	0	0	0	-1	0	1	0	30
	0	0	1	0	0	0	-1	0	1	20

Canonization

$$R_2 \leftarrow R_2/0.02$$

	Z	x_1	x_2	s_1	s_2	e_1	e_2	a_1	a_2	b
	1	0	-1M	0	0	-300	M	1M	0	-20M
	0	0	1	58.8235	0	1.47059	0	-1.47059	0	14.7059
	0	0	0.02	0	1	0.02	0	-0.02	0	0.4
	0	1	0	0	0	-1	0	1	0	30
	0	0	1	0	0	0	-1	0	1	20

$$R_1 \leftarrow R_1 + 1000200.00R_2$$

Z	x_1	x_2	s_1	s_2	e_1	e_2	a_1	a_2	b
1	0	0	58.8M	0	1.47M	M	-0.471M	0	-5.28M
0	0	1	58.8235	0	1.47059	0	-1.47059	0	14.7059
0	0	0.02	0	1	0.02	0	-0.02	0	0.4
0	1	0	0	0	-1	0	1	0	30
0	0	1	0	0	0	-1	0	1	20

$$R_3 \leftarrow R_3 + -0.02R_2$$

Z	x_1	x_2	s_1	s_2	e_1	e_2	a_1	a_2	b
1	0	0	58.8M	0	1.47M	M	-0.471M	0	-5.28M
0	0	1	58.8235	0	1.47059	0	-1.47059	0	14.7059
0	0	0	-1.17647	1	-0.00941176	0	0.00941176	0	0.105882
0	1	0	0	0	-1	0	1	0	30
0	0	1	0	0	0	-1	0	1	20

$$R_4 \leftarrow R_4 + -0.00R_2$$

Z	x_1	x_2	s_1	s_2	e_1	e_2	a_1	a_2	b
1	0	0	58.8M	0	1.47M	M	-0.471M	0	-5.28M
0	0	1	58.8235	0	1.47059	0	-1.47059	0	14.7059
0	0	0	-1.17647	1	-0.00941176	0	0.00941176	0	0.105882
0	1	0	0	0	-1	0	1	0	30
0	0	1	0	0	0	-1	0	1	20

$$R_5 \leftarrow R_5 + -1.00R_2$$

Z	x_1	x_2	s_1	s_2	e_1	e_2	a_1	a_2	b
1	0	0	58.8M	0	1.47M	M	-0.471M	0	-5.28M
0	0	1	58.8235	0	1.47059	0	-1.47059	0	14.7059
0	0	0	-1.17647	1	-0.00941176	0	0.00941176	0	0.105882
0	1	0	0	0	-1	0	1	0	30
0	0	0	-58.8235	0	-1.47059	-1	1.47059	1	5.29412

Pivot Result

Z	x_1	x_2	s_1	s_2	e_1	e_2	a_1	a_2	b
1	0	0	58.8M	0	1.47M	M	-0.471M	0	-5.28M
0	0	1	58.8235	0	1.47059	0	-1.47059	0	14.7059
0	0	0	-1.17647	1	-0.00941176	0	0.00941176	0	0.105882
0	1	0	0	0	-1	0	1	0	30
0	0	0	-58.8235	0	-1.47059	-1	1.47059	1	5.29412

Non-Feasible

The artificial variable a_2 is in the base. Therefore, a BFS (basic-feasible-solution) does not exist for this problem.